Engine Model(s): PT6A

79-20-01

EXTERNAL OIL TUBES - DESCRIPTION AND OPERATION

1. Description and Operation

A. Pressure System

Pressure oil from the filter outlet is delivered to the engine bearing areas, the accessory and reduction gearboxes and to the oil-to-fuel heater circuit through internal cored passages and internal and external oil transfer tubes. Pressure oil is delivered to the No. 1 bearing and the accessory gearbox via cored passages and internal transfer tubes, while oil to Nos. 2, 3 and 4 bearings and reduction gearbox is passed through a cored passage to an elbow fitting at the 5 o'clock position on the compressor inlet case, then through an external transfer tube to a similar elbow fitting at the front end of the reduction gearbox. Oil is then delivered to the various areas within the gearbox by cored passages and transfer tubes. Oil to the No. 2 bearing is tapped off the external transfer tube and directed to the bearing area through internal piping.

B. Scavenge System

Scavenge oil from the No. 2 bearing area is directed through internal tubes to a port at the 7 o'clock position on the gas generator case. The flanged No. 2 bearing scavenge tube directs oil rearward to a boss at 6 o'clock on the compressor inlet case. The oil is then passed through internal transfer tubes to the front element of the internal scavenge pump which discharges the scavenged oil into the accessory gearbox sump.

Oil from the reduction gearbox and No. 3 and 4 bearing areas is returned through twin external tubes to respective scavenge pumps at the rear of the engine. The stainless steel scavenge tubes are connected to the respective ports on the reduction gearbox through short transfer tubes and a common twin-port adapter bolted to the front case of the gearbox. The scavenge tubes are supported by a flange-mounted coupling on flange C. The rear ends of the tubes fit into elbow extensions of the front and rear elements of the external scavenge pump. The No. 3 and 4 bearing oil is scavenged by the front element and discharged into the accessory gearbox sump, while the reduction gearbox oil is scavenged by the rear element which returns the oil to the oil tank via an airframe-mounted cooler (Ref. Aircraft Maintenance Manual).

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